

Paper Manager pack — π_0 (Sep 28)

Paper: π_0 : A Vision-Language-Action Flow Model for General Robot Control

arXiv: <https://arxiv.org/abs/2410.24164> · RSS 2025

Blog: <https://physicalintelligence.company/blog/pi0>

PDF: /workspace/papers/pi0-2410.24164.pdf

Session: Sep 28 · due 9:30 AM · **You = Archaeologist (seat 1)**

Type: VLA Architectures

Sources: Summarizer + Senior Research (/workspace/pi0-quiet/DELIVERABLE.md)

1. Paper summary

Robot foundation VLA: **PaliGemma (~3B) + ~300M action expert** trained with **OT-CFM** to emit continuous action chunks **H=50** at up to **50 Hz**. Pretrain → post-train on **~10k hours** across **7 robots / 68 tasks** (+ OXE/Bridge/DROID). Beats OpenVLA/Octo out-of-box; FT beats ACT/DP; pretrain+FT unlocks multi-minute household skills.

Club bridge: SigLIP → PaliGemma → π_0 → OpenPI → $\pi_{0.5}$ → RTC → BEHAVIOR'25 → '26 baselines.

2. Keywords / stack snapshot

Item	Value
Backbone	PaliGemma 3B + ~300M FM action expert (total ~3.3B)
Objective	OT-CFM; τ -shifted Beta(1.5,1); 10 Euler steps
Chunk	H=50 train; exec 16@20Hz / 25@50Hz (no temporal ensembling)
Latency	~73 ms on-board RTX4090 (Table I)
Data	~10k h proprietary teleop; mixture weight $n^{0.43}$

3. Motion / control spine

DP/ACT **chunk pattern** with **DDPM** → **OT-FM** (n=10). OpenVLA fails partly from no chunking. Open-loop seam → later **RTC** soft-inpaint. BEHAVIOR keeps this substrate with correlated ϵ , N=15, exec26/keep4.

Orthogonal knobs: H (train horizon) · Ta (open-loop commit) · NFE (Euler steps) · async overlap · Σ (noise correlation).

NOT a classical planner — generative continuous chunk controller.

4. Role packs

Stakeholder (18+3) — seat 2

Reference 2025–26 VLA template. Start from OpenPI π_{i0} / π_{i05_base} + FT. Budget multi-cam + ~73–86ms class GPU. Risks: unreproducible 10k h; open-loop delay; not locomotion-universal.

Scientific Reviewer (8+3) — seat 4

Strengths: integrative architecture+data+systems; right inductive bias vs AR VLAs; ~10k h scale; ablations (small/parity/scratch); honest difficulty.

Weaknesses: figure-only floats; π_0 -small confounds; baseline compute unmatched; n=10 little variance reported; external validity limited (PI robots).

Verdict: Strong systems+empirical RSS; shifted field toward flow/diffusion chunks.

Empiricist (11+3) — seat 5

- H1:** mid commit (~0.5–1s / H≈16–32) beats too-short and full open-loop H=50
- H2:** success saturates ~5–10 NFE
- H3:** FT from π >> zero-shot/scratch on multi-stage tasks
- H5:** FM lower NFE/latency than DP at matched backbone

Priority	Experiment
P0	$H \in \{8, 16, 25, 50\}$; $NFE \in \{1, 2, 4, 5, 10, 20\}$; FT vs zero-shot/scratch
P1	Cross-embodiment; FM vs DP head
P2	τ -schedule; open-loop commit; VLM init vs scratch expert
Defer	Full ~10k-h pretrain; full laundry/box suite

Budget: P0 on public OpenPI ckpts = 1–4×GPU days (not weeks). Full pretrain irreproducible. Env: /workspace/pi0-quiet/env_outline/.

Archaeologist (11+3) — ****YOU (seat 1)**** — HIGH PRIORITY

Influential priors (CLEAR)

Prior	Role
SigLIP	Vision tower lineage
PaliGemma (2407.07726)	VLM init
Lipman FM (2210.02747)	OT-CFM objective (π_0 cites)
Rectified Flow	Concurrent straight-path flows
Diffusion Policy (2303.04137)	Generative action chunks (DDPM)
ACT	Parallel chunking / idle robustness
Transfusion	MoE-style dual-stream inspiration
RT-2 / OpenVLA	AR VLA foils
OXE / Bridge / DROID	Open data mixture ingredients
SayCan	Language → robot planning foil

Pivot this paper locks: FM VLA + VLM init + action expert + 50 Hz chunks at ~10k h scale.

CLEAR subsequent

Work	Link
OpenPI (2025-02)	Open release / ecosystem
FAST / π_0 -FAST (2501.09747)	Discrete tokenization branch
$\pi_{0.5}$ (2504.16054)	FM **retained** at post-train (blogs saying FAST replaced FM = **false**)
RTC (2506.07339)	Soft-inpaint async chunks
BEHAVIOR’25 1st (2512.06951)	On Pi0.5; q~0.26
BEHAVIOR’26	Baselines include $\pi_{0.5}$

Genealogy slide:

SigLIP → PaliGemma → π_0 → OpenPI → (FAST || $\pi_{0.5}$) → RTC → BEHAVIOR’25 forks → BEHAVIOR’26 baselines.

Visionary (11+3) — seat 3

2026 BEHAVIOR: FM+chunk = default control prior; 2025 winners specialized noise/overlap not the head. Hierarchy/stage memory next bottleneck. Deploy with RTC. Start pi05_base; port correlated Σ ,

multi-sample FM, soft inpaint, stages. Durable idea: **VLM semantic prior + generative continuous chunks**. Skip Discord/Drive.

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Template VLA + OpenPI start
Reviewer	8+3	Strengths / private-data limits
Empiricist	11+3	H / NFE / FT grid
Archaeologist	**11+3**	**Priors + genealogy timeline (HIGH)**
Visionary	11+3	2026 BEHAVIOR defaults

Spine: PaliGemma + OT-CFM chunks (DDPM → flow). PDF for Mac. Include all five roles for GitHub.

Paths

Artifact	Path
This pack	`/workspace/pi0-quiet/KANTA_PACK.md`
Senior Research	`/workspace/pi0-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/pi0-2410.24164-summary.md`
Paper PDF	`/workspace/papers/pi0-2410.24164.pdf`